



METRO FRESH

Shrink Loss Analysis

A Data-Driven Investigation into Inventory Shrinkage,
Loss Attribution & Strategic Recovery Pathways

BY: ESMERALDA OSEI

Metro Fresh Enterprise

A multi-format convenience retail chain operating across diverse locations, stocking a wide range of goods across 15+ categories and serving thousands of daily transactions.

\$8.88M+

Total Revenue

50+

Store Locations

15+

Product Categories

4

Store Formats

Store Network

Operations span Urban (48%), Suburban (35%), Highway (10%), and Rural (7%) formats — each presenting unique risk profiles.

Data Scope

Analysis covers inventory records, shrink transactions, shift logs, cause attribution, and category-level KPIs across the full period.

PROBLEM STATEMENT

Shrink & Loss Context

Metro Fresh is experiencing persistent inventory shrinkage across its store network, representing a material financial drain against total revenue. Despite recording over \$8.88M in sales, shrink losses remain substantial and unevenly distributed — with no consistent, data-driven mechanism to identify, attribute, or reduce these losses.

Shrink Exceeds Target

Multiple high-value categories consistently breach shrink rate targets with no corrective action triggered.

Unclear Loss Attribution

Losses split across expiry, damage, theft, and staff consumption with no prioritised response framework.

Inventory Discrepancies

Large gaps between expected and actual SKU-level inventory signal systemic record-keeping or process failures.

Store-Level Concentration

A small subset of stores drives a disproportionate share of total shrink, indicating localised control gaps.

ANALYTICAL OBJECTIVES

Operational Research Framework

This analysis was designed to answer six core operational questions to drive strategic mitigation:

01. WHERE

Where is shrink happening?

Identify which stores, formats, and categories carry the highest loss burden.

02. WHY

Why is shrink happening?

Attribute losses by root cause: expiry, damage, theft, employee consumption.

03. WHEN

When does shrink peak?

Analyse daily and monthly trends to detect temporal patterns and spikes.

04. TARGETS

Are targets being met?

Compare actual vs target shrink rates across all product categories.

05. ACCURACY

Is inventory accurately tracked?

Assess record accuracy and variance between expected and actual stock levels.

06. ACTION

What actions can reduce loss?

Translate data findings into prioritised, actionable recommendations.

EXECUTIVE REPORT OVERVIEW

MetroFresh Executive Shrink & Performance Report

Shrinkage & Inventory Accuracy Analysis · Nov 2024 – Apr 2025

Executive Dashboard

Operational Analysis
Dashboard

Month

All

Store Format

All

state

All

Category

All

\$8.88M

Total Revenue

42.50%

Gross Profit %

\$217.12K

Total Shrink Value

2.44%

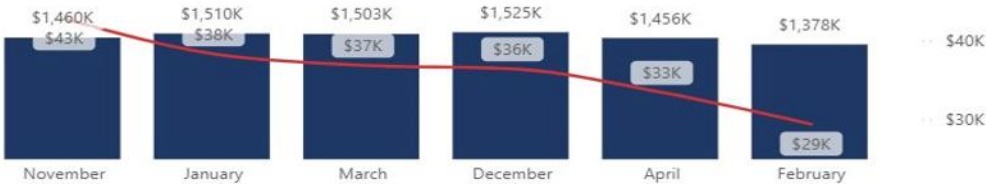
Shrink Rate %

97.01%

Inventory Record Accuracy

Monthly Revenue vs Shrink Loss Trend

● Total Revenue ● Total Shrink Value Absolute



Shrink Loss Distribution by Store Format

● urban ● suburban ● highway ● rural



Inventory Record Accuracy vs Target

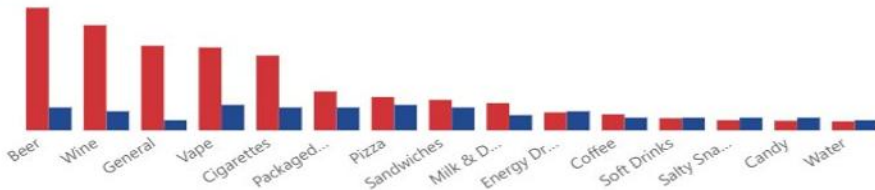


Top 5 Stores by Shrink Loss



Actual vs Target Shrink Rate by Category

● Shrink Rate % Display ● Target Shrink Rate



KPI Cards — Headline Performance Metrics

KPI



Total Revenue - \$8.88M



Gross Profit Margin-
42.50%



Total Shrink Value-
\$217.12K



Shrink Rate- 2.44%



Inventory Record Accuracy-
97.01%

KEY FINDINGS

Revenue & Profit: \$8.88M total revenue with **42.50%** gross profit — the financial baseline against which all shrink impact is measured.

Total Shrink Value (\$217.12K): Red conditional formatting is a direct visual alert. Executives immediately see that shrink has exceeded acceptable levels.

Shrink Rate (2.44%): 1.04pp above the 1.40% chain target and above the 1.8% industry benchmark. This confirms this is the **core performance problem** requiring prioritised action.

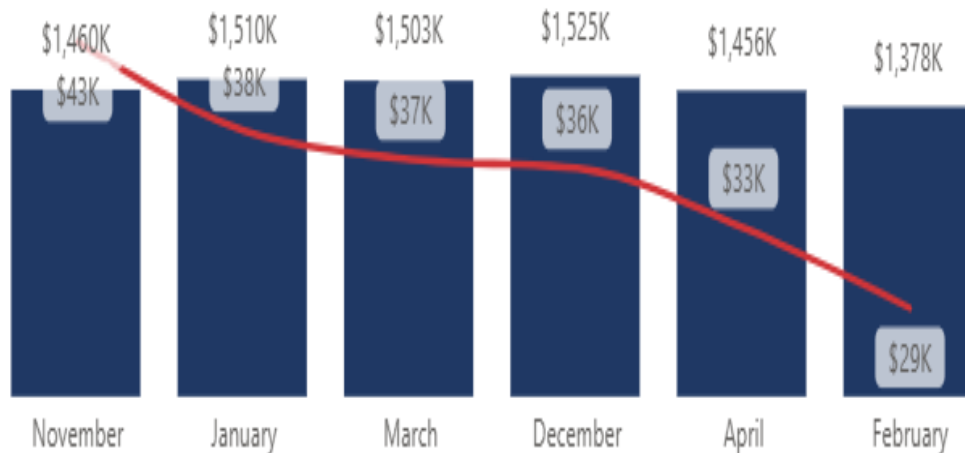
Inventory Accuracy (green — 97.01%): Green formatting confirms above the 92% target — one of the few **positive signals** on the dashboard.

Monthly Revenue vs Shrink Loss Trend

TREND

Monthly Revenue vs Shrink Loss Trend

● Total Revenue ● Total Shrink Value Absolute



KEY FINDINGS

Revenue Stability: Revenue ranged between **£1.37M and £1.51M** — relatively stable across the period.

Shrink Reduction: Shrink loss fell from **£43K (Nov) to £29K (Feb)** — a 33% reduction over 6 months.

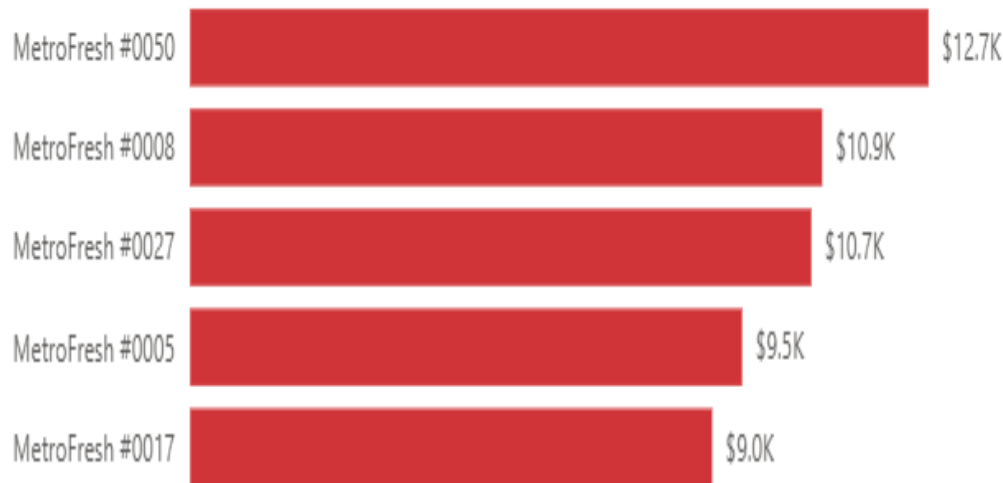
Trend Analysis: The downward shrink trend is a **positive signal** but has not yet fully stabilized.

Risk Factors: Peak shrink in **November** aligns with high footfall, raising theft and expiry risk in busy periods.

Top 5 Stores by Shrink Loss

STORES

Top 5 Stores by Shrink Loss



KEY FINDINGS

Top Performer: **MetroFresh #0050** is the **highest-loss store** at £12.7K — 41% more than the 5th-ranked store.

Concentration: The top 5 stores collectively drive a significant concentration of total network shrink.

Urgent Review: **Stores #0008** and **#0027** are closely matched at £10.9K and £10.7K — both require urgent review.

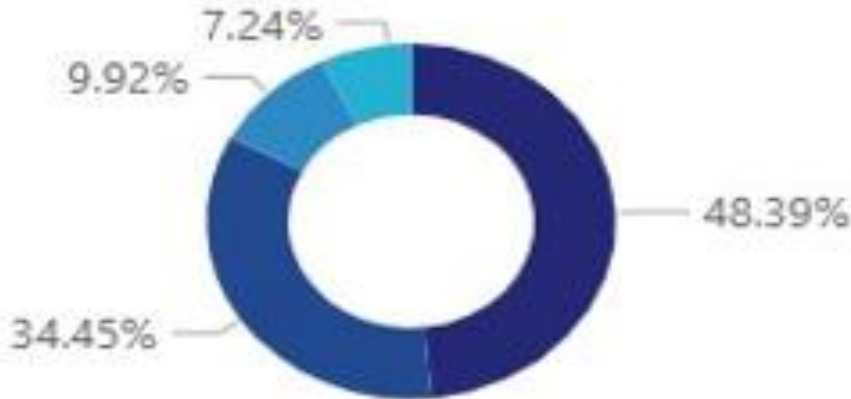
Insight: Loss concentration suggests controllable store-level process failures rather than systemic issues.

Shrink Loss Distribution by Store Format

FORMAT

Shrink Loss Distribution by Store Format

● urban ● suburban ● highway ● rural



ANALYSIS SUMMARY

Urban Dominance: **Urban stores** account for **48.39%** of all shrink — the single largest segment in the network.

Suburban Volume: **Suburban stores** contribute **34.45%**, reflecting higher transaction volumes in residential areas.

Minor Segments: **Highway (9.92%)** and **rural (7.24%)** formats show lower absolute loss but may have lower oversight.

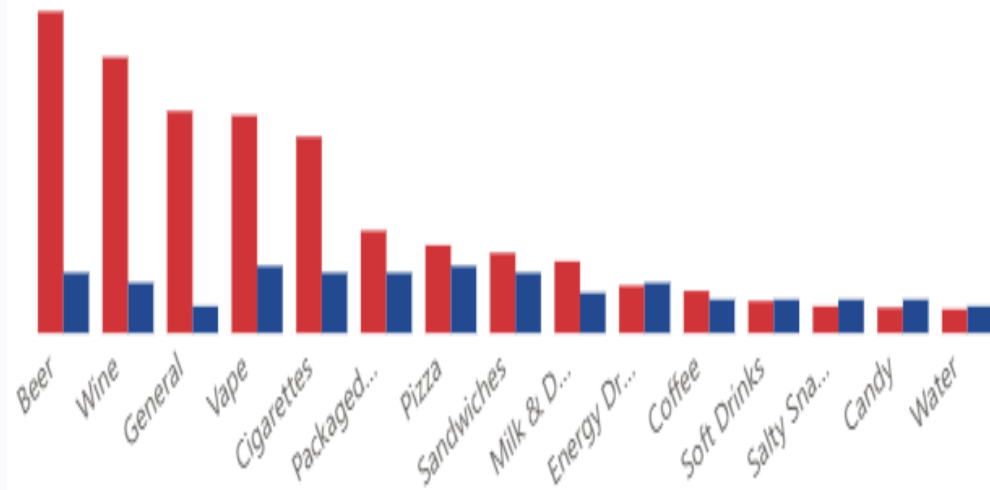
Action Required: Urban shrink concentration warrants dedicated loss prevention resources in city-format locations.

Actual vs Target Shrink Rate by Category

CATEGORY

Actual vs Target Shrink Rate by Category

● Shrink Rate % Display ● Target Shrink Rate



KEY FINDINGS

High Theft Risk: **Beer** and **Wine** show the largest gap above target — both are high-value, high-theft categories.

Significant Breaches: General merchandise, Vape, and Cigarettes also significantly exceed their shrink rate targets.

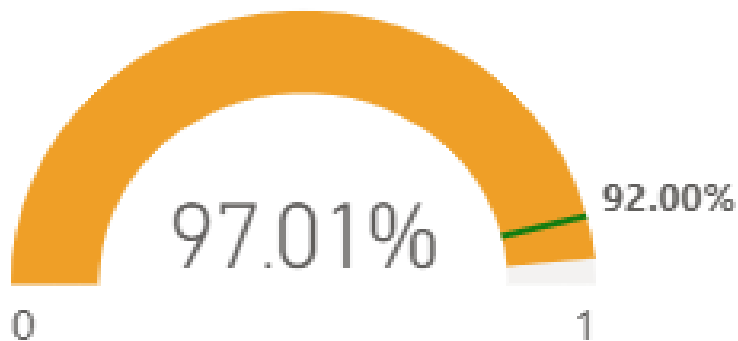
Low Risk Areas: Water, Candy, and Salty Snacks are closest to or within target — lowest risk categories.

Network Health: 5 of 15 categories show persistent target breaches requiring category-specific interventions.

Inventory Record Accuracy vs Target

ACCURACY

Inventory Record Accuracy vs Target



KEY FINDINGS

Performance: Current inventory record accuracy stands at **97.01%** — exceeding the 92% target by 5 points.

Reliability: One of the few strongly positive KPIs, suggesting the counting and recording process is reliable.

Confidence: High accuracy supports confidence that reported shrink figures reflect real losses, not data errors.

Maintenance: Maintaining this level requires consistent cycle counting discipline, especially in high-shrink stores.

OPERATIONAL REPORT OVERVIEW

Operational Report Analysis

Inventory Reconciliation · Shrink Attribution · Shift & Anomaly Analysis · Nov 2024 – Apr 2025

Executive Dashboard

Operational Analysis
Dashboard

Stores

All

Categories

All

Theft Risk Tier

All

Year

All

1M

Expected Inventory

824K

Actual Inventory

339K

Expected Inventory

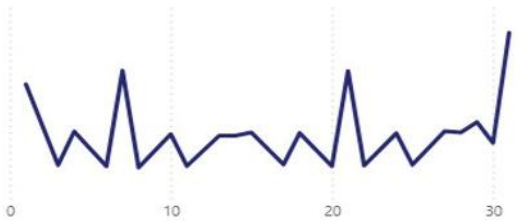
Negative Inventory Adjustments by Cause



Shrink Value by Shift Classification



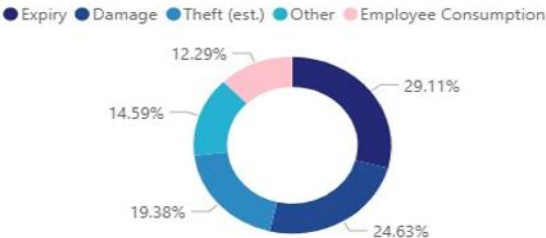
Daily Shrink Rate Trend



SKU Level Expected vs Actual Inventory

Store Name	Expected Inventory	Actual Inventory	Expected vs Actual Variance	Total Shrink Units	Total Shrink Value Absolute
MetroFresh #0045	37480	6763	-30717	-115	\$1,024
MetroFresh #0002	34139	6527	-27612	-188	\$1,589
MetroFresh #0025	32375	7004	-25371	-118	\$940
MetroFresh #0048	31285	7008	-24277	-154	\$1,360
MetroFresh #0029	28452	6893	-21559	-185	\$1,842
MetroFresh #0009	27553	7014	-20539	-268	\$2,342
MetroFresh #0015	26707	6590	-20117	-157	\$1,370
MetroFresh #0023	32950	12964	-19986	-188	\$1,829
MetroFresh #0046	33065	13140	-19925	-259	\$2,121

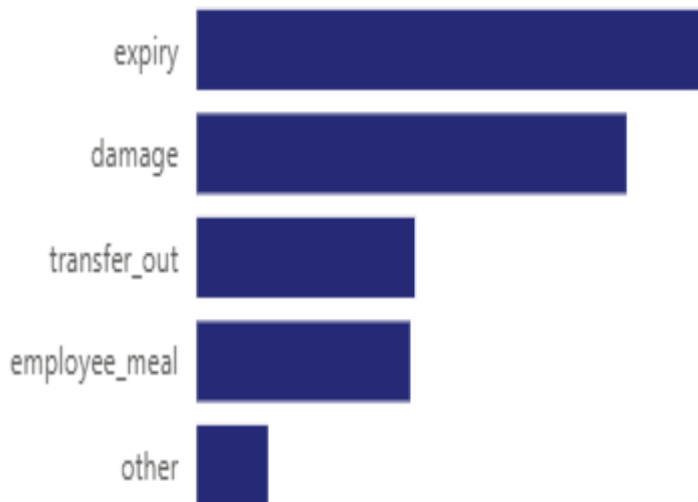
Shrink Loss Attribution by Cause



Negative Inventory Adjustments by Cause

ADJUSTMENTS

Negative Inventory Adjustments by Cause



KEY FINDINGS

Expiry: Generates the **highest** volume of negative adjustments — far exceeding all other causes.

Damage: The **second largest** adjustment cause, consistent with the attribution chart findings.

Process Control: Transfer out and employee meal adjustments are mid-range — both controllable with process changes.

Reclassification: 'Other' adjustments are the smallest bucket but should be investigated and reclassified where possible.

Daily Shrink Rate Trend

DAILY

KEY FINDINGS

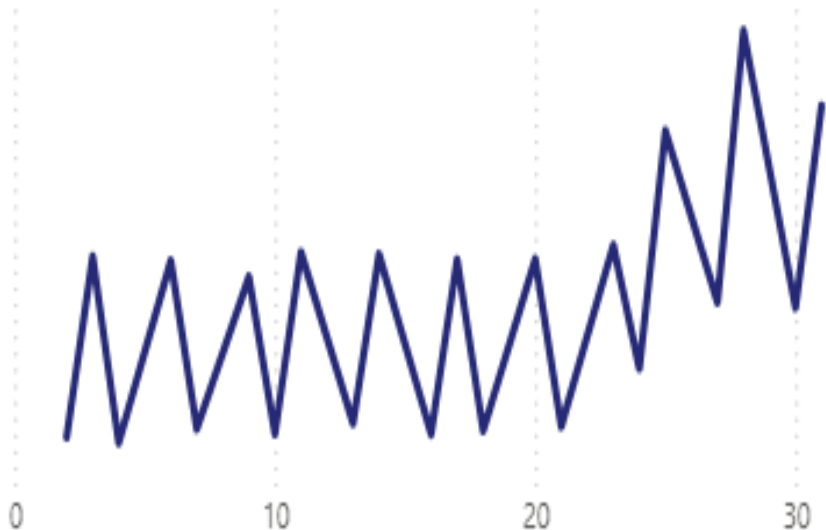
Stable Trend: Shrink rate is broadly stable across **days 1-25**, oscillating within a consistent range.

Period Spike: A sharp spike is visible around **day 30** — suggesting an end-of-period anomaly or reporting lag.

Cyclical Peaks: Secondary spikes around days 8 and 20 may correspond to product expiry batches or deliveries.

Alert Threshold: Daily alerts should trigger when shrink rate exceeds the observed baseline by more than 1.5x.

Daily Shrink Rate Trend

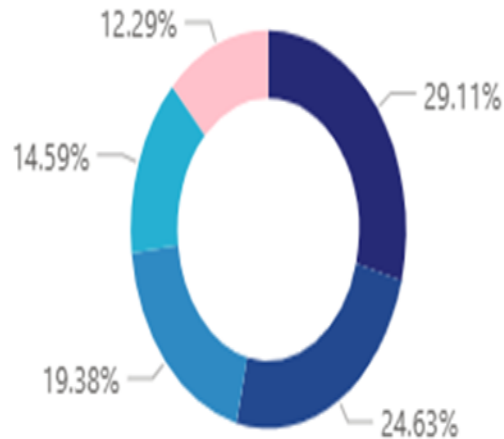


Shrink Loss Attribution by Cause

CAUSES

Shrink Loss Attribution by Cause

● Expiry ● Damage ● Theft (est.) ● Other ● Employee Consumption



KEY FINDINGS

Expiry: Leading cause at **29.11%** — indicating stock rotation and ordering process failures.

Damage: Accounts for **24.63%** — the second largest contributor, linked to handling and storage.

Theft (est.): is the third largest at **19.38%**.

Minor Causes: Other unclassified shrink sits at **14.59%**.

SKU Level Expected vs Actual Inventory

SKU

SKU Level Expected vs Actual Inventory

Store Name	Expected Inventory	Actual Inventory	Expected vs Actual Variance	Total Shrink Units	Total Shrink Value Absolute
MetroFresh #0045	37480	6763	-30717	-115	\$1,024
MetroFresh #0002	34139	6527	-27612	-188	\$1,589
MetroFresh #0025	32375	7004	-25371	-118	\$940
MetroFresh #0048	31285	7008	-24277	-154	\$1,360
MetroFresh #0029	28452	6893	-21559	-185	\$1,842
MetroFresh #0009	27553	7014	-20539	-268	\$2,342
MetroFresh #0015	26707	6590	-20117	-157	\$1,370
MetroFresh #0023	32950	12964	-19986	-188	\$1,829
MetroFresh #0046	33065	13140	-19925	-259	\$2,121

KEY FINDINGS

Max Variance: **MetroFresh #0045** has the largest variance: **-30,717 units** between expected and actual inventory.

Systemic Loss: Negative variances are consistently large across the top stores, indicating systemic loss issues.

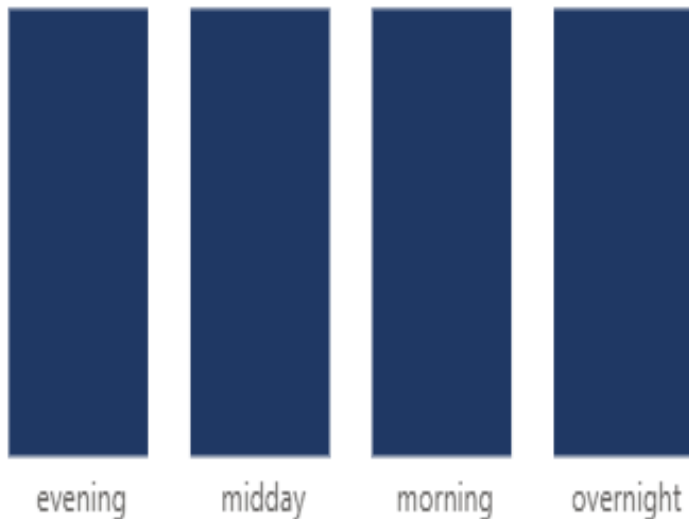
Highest Shrink: **MetroFresh #0009** shows the **highest shrink value (£2,342)** despite not having the worst unit gap.

Targeted Focus: SKU-level monitoring should focus on stores with both large unit gaps and high £ shrink values.

Shrink Value by Shift Classification

SHIFTS

Shrink Value by Shift Classification



KEY FINDINGS

Uniform Distribution: Shrink is distributed almost **equally** across evening, midday, morning, and overnight shifts.

No Vulnerability: No single shift carries a disproportionate share, ruling out a single-window vulnerability.

Overnight Risk: The overnight shift has fewer staff — making equivalent shrink there more operationally significant.

Data Correlation: Shift analysis should be cross-referenced with theft and employee consumption data for clarity.

INSIGHTS & RECOMMENDATIONS

HIGH

Address Expiry Losses

Review ordering patterns for Beer, Wine & Vape. Implement FIFO strictly, add markdown triggers 48 hrs before expiry.

HIGH

Intervene in Top 5 Stores

Deploy dedicated loss prevention audits at #0050, #0008 & #0027. Set store-specific shrink KPIs with weekly review.

HIGH

Strengthen Urban Controls

Urban stores carry 48% of shrink. Increase CCTV coverage, cycle counting frequency, and shift handover checks.

MED

Reclassify 'Other' Shrink

19% unclassified shrink limits response capability. Introduce mandatory sub-category coding at point of adjustment.

MED

Investigate Day 30 Spike

The daily trend spike warrants root cause investigation — likely an end-of-period reporting or delivery event.

LOW

Sustain Accuracy Rate

Inventory accuracy at 97% is a strong asset. Formalise the counting process to protect this KPI long-term.



METRO FRESH

Shrink Loss Analysis

Insights generated from November 2024 – April 2025 retail operations data.

THANK YOU.